

Computer Ports — Detailed Lecture Notes

JKSSB Computer Awareness

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16.1 What a port is

Every device that connects to a computer does so through a **port**. A port is a socket or interface on the machine into which a cable or a device is plugged. It lets data and, in many cases, electric power pass between the computer and the attached device. Ports are not all alike: they differ in their shape, in the type of signal they carry (digital or analog), and in the job they are meant to do.

The common physical ports found on a personal computer are USB (including USB-C), HDMI, DisplayPort, VGA, Ethernet through an RJ-45 connector, the 3.5 mm audio jack, and SATA. Knowing which port suits which device is the central skill of this lesson.

Definition (Port). *A socket or interface on a device where a cable or another device is connected, allowing data (and often power) to flow between the computer and the attached device.*

Term	Meaning in this lesson
Port	The socket on the device (a USB port, an HDMI port).
Connector	The plug end of a cable that fits the port.
Cable	The wire that joins two connectors.
Physical port	A hardware socket on the machine.
Logical port	A software number that names a network service.

A first view of the ports covered in this lesson, with their signal type and their normal use, is given below. The rest of the lesson explains each one in turn.

Port	Signal	Typical use
USB / USB-C	Digital (data + power)	Keyboard, mouse, pen drive, phone
HDMI	Digital (video + audio)	TV, monitor, projector
DisplayPort	Digital (video + audio)	PC monitors, graphics cards
VGA	Analog (video only)	Older monitors and projectors
RJ-45 (Ethernet)	Digital (network)	Wired LAN cable, router, switch
3.5 mm audio jack	Analog (audio)	Headphones, speakers, microphone
SATA / eSATA	Digital (storage)	Internal and external drives

16.2 The two meanings of “port”

The word **port** has two entirely different meanings in computing, and the exam frequently tests the difference. A **physical port** is a piece of hardware — a socket such as a USB port, an HDMI port, or an RJ-45 network port. A **logical port**, also called a network port, is a software number that identifies a particular service running on a computer during network communication.

A web server, for example, listens on logical port 80 for ordinary web requests and on port 443 for secure web requests. The physical network cable, meanwhile, is plugged into an RJ-45 port on the network card. Thus one machine uses both meanings at the same time without any contradiction: the socket is hardware, and the port number is software.

Definition (Logical (network) port). *A number, in the range 0 to 65535, that identifies a specific service or process on a computer for the purpose of network communication. It is not a physical socket.*

Example 16.2.1. *An item states, “The monitor is connected to logical port 80.” This is nonsense. Port 80 is a logical port used by web (HTTP) traffic, not a socket for a monitor. A monitor connects to a physical video port such as HDMI, DisplayPort, or VGA. The wrong statement has mixed the two meanings of the word port.*

A few well-known logical port numbers are worth remembering, because the exam sometimes asks which service uses a given port. The list below is illustrative; the concept being tested is that these are *numbers*, not sockets.

Service	Logical port	Full form
HTTP	80	HyperText Transfer Protocol
HTTPS	443	HTTP Secure
FTP	21	File Transfer Protocol
SMTP	25	Simple Mail Transfer Protocol
DNS	53	Domain Name System

16.3 USB: the Universal Serial Bus

USB stands for **Universal Serial Bus**. It is a standard interface for connecting a wide range of peripherals to a computer. Two of its best-known features are **plug and play** and **hot swapping**: a device can be attached or removed while the computer is running, without switching the machine off. A USB connection carries both **data** and **power**, which is why many small devices can be used or charged directly from a USB port.

Typical USB devices include the keyboard, the mouse, a printer, a pen drive (flash drive), an external hard disk, a webcam, and a smartphone. The connector at the end of a USB cable comes in several shapes:

- **USB Type-A**: the familiar rectangular connector found on most PCs.
- **USB Type-B**: often used for printers and scanners.
- **Mini-USB** and **Micro-USB**: smaller connectors used on older phones and devices.
- **USB Type-C**: a modern, small, reversible connector.

USB-C, or USB Type-C, deserves special attention. It is a *connector type*, not a separate bus. It is **reversible**, so it can be plugged in either way up, and a single USB-C connection can carry data, power, and video together. It is now common on smartphones, laptops, external SSDs, and docking stations. The key exam point is precision: USB is the standard, USB-C is one connector shape for that standard, and a USB port is the socket on the machine.

The USB standard itself has developed over time through versions such as USB 1.1, USB 2.0, USB 3.0 (and its later 3.1 and 3.2 refinements), and USB4. Each newer version generally offers a higher data-transfer rate, and the version is independent of the connector shape: a Type-A or Type-C socket can appear on a machine that supports different USB versions. For exam purposes, the safe points are the meaning of the abbreviation, the plug-and-play and hot-swap behaviour, the combined data-and-power carrying, and the reversibility of USB-C.

16.4 Video ports: HDMI, DisplayPort, and VGA

Displays are attached through video ports. Three names dominate the syllabus.

HDMI stands for **High-Definition Multimedia Interface**. It carries **digital video and digital audio** over a single cable, which is why one HDMI cable can replace a separate video cable and a separate audio cable. It is very common on televisions, monitors, projectors, set-top boxes, and game consoles.

DisplayPort is also a **digital** audio/video interface, but it is designed mainly for computers and monitors. It appears on graphics cards, laptops, and desktop monitors, and it supports high refresh rates and multiple monitors through daisy-chaining. **Mini DisplayPort** is a smaller variant of the same interface.

Point	HDMI	DisplayPort
Signal	Digital video + audio	Digital video + audio
Main use	TV, home theatre, monitor	PC monitors, graphics cards
Note	Very common on TVs	Common on PCs; multi-monitor

Both HDMI and DisplayPort are digital. The right choice in practice usually depends on which sockets the devices offer, not on one being inherently better than the other.

VGA stands for **Video Graphics Array**. It is an **analog** video interface: it carries video only and no audio. It uses a 15-pin D-sub connector, often coloured blue and held in place by two screws. VGA is still seen on older monitors and projectors but has largely been replaced by HDMI and DisplayPort.

Definition (Digital vs analog video). *Digital video ports (HDMI, DisplayPort) send the picture as discrete numbers and also carry audio; the analog port (VGA) sends a continuously varying signal and carries video only.*

Example 16.4.1. *An item reads, “VGA carries both video and audio.” This is false. VGA is an analog video-only interface with no audio channel. HDMI and DisplayPort, by contrast, carry digital video together with digital audio. The wrong option has transferred the audio capability of the digital ports onto VGA.*

16.5 Ethernet and the RJ-45 connector

Ethernet is the standard for **wired** computer networking, and it is commonly associated with the IEEE 802.3 family of standards. The physical socket used for a wired network connection is

the **RJ-45** connector. RJ-45 stands for **Registered Jack 45**, and it has eight pins, used with twisted-pair cable such as Cat5e or Cat6.

Typical devices attached through an RJ-45 port are a router, a switch, a modem, and the network card of a desktop computer. A near-neighbour that causes confusion is **RJ-11**, the smaller telephone jack. RJ-45 is the network connector with eight pins; RJ-11 is the telephone connector and is smaller. The two look similar but serve different purposes.

16.6 Audio jacks

The common audio socket on a personal computer is the **3.5 mm audio jack**. It connects headphones, earphones, speakers, and microphones. The plug wiring has standard forms: **TRS** (Tip-Ring-Sleeve) carries stereo audio, while **TRRS** (Tip-Ring-Ring-Sleeve) adds a microphone channel, which is why a phone headset with a microphone uses a TRRS plug. On desktop PCs a colour convention is often followed: green for audio output, such as headphones or speakers, and pink for a microphone. Some equipment uses 2.5 mm or 6.35 mm jacks instead of the 3.5 mm size.

16.7 Storage interfaces: SATA and eSATA

SATA stands for **Serial Advanced Technology Attachment**. It is the interface that connects **internal storage** — a hard disk drive, a solid-state drive, or an optical drive — to the motherboard. The SATA data cable is thin and has seven pins, and a separate power connector feeds the drive. **eSATA** stands for **external SATA**, and it is used to connect an external drive. It is important to classify eSATA correctly: it is a *storage interface*, not a general-purpose peripheral port like USB. SATA is about attaching drives, whereas USB, HDMI, and the audio jack are about attaching other kinds of peripherals and displays.

16.8 Classifying a port: four questions to ask

When a question shows a port or names it, four questions settle almost every case. First, *is it a physical port or a logical one?* A socket is physical; a number such as 80 or 443 is logical. Second, *what kind of signal does it carry?* HDMI and DisplayPort are digital audio/video, VGA is analog video, the 3.5 mm jack is analog audio, and RJ-45 is a digital network link. Third, *what class of device does it attach?* Displays go to video ports, drives go to SATA, network equipment goes to RJ-45, and general peripherals go to USB. Fourth, *which near-neighbour is it often confused with?* USB-C is confused with USB generally; RJ-45 is confused with RJ-11; VGA is confused with HDMI and DisplayPort.

Definition (Port class). *A way of grouping ports by what they attach: peripheral ports (USB), display ports (HDMI, DisplayPort, VGA), the network port (RJ-45), audio ports (3.5 mm jack), and storage interfaces (SATA, eSATA).*

Two ports are easily confused with a near-neighbour. The table below keeps the distinctions in one place.

Port	Nearest confusion	The distinction
USB-C	USB (the standard)	USB-C is a connector shape; USB is the bus.
HDMI	DisplayPort	Both digital A/V; HDMI is common on TVs, DisplayPort on PCs.
VGA	HDMI / DisplayPort	VGA is analog, video-only; the others are digital with audio.
RJ-45	RJ-11	RJ-45 is the 8-pin network jack; RJ-11 is the smaller phone jack.
SATA	USB	SATA attaches drives (storage); USB is a general peripheral port.
Logical port	Physical port	Logical port is a number; physical port is a socket.

16.9 Exam retrieval and common confusions

The following distinctions prevent most errors on this topic.

1. A **physical port** is a hardware socket; a **logical (network) port** is a software number, such as HTTP 80 or HTTPS 443. Never mix the two meanings of “port”.
2. **VGA is analog** and video-only; **HDMI and DisplayPort are digital** and carry both video and audio.
3. **RJ-45** (network, eight pins) is not **RJ-11** (the smaller telephone jack).
4. **USB-C is a connector type** and is reversible; “USB” is the standard, and a USB port is the socket.
5. **SATA is a storage interface**; do not call it a display or network port.
6. Say the term precisely: a *port* is the socket, a *connector* is the plug, and a *cable* is the wire. This is the same discipline as file *format* vs file *extension* (TRM-09).

Example 16.9.1. *An item asks which port carries digital audio and digital video over one cable. Name the digital audio/video ports: HDMI and DisplayPort both qualify, whereas VGA is analog video-only, RJ-45 is networking, and SATA is storage. If the stem prefers the port common on television sets, choose HDMI; if it stresses PC monitors and multi-monitor setups, choose DisplayPort.*

The full forms below are the ones most likely to be asked directly, so they should be memorised exactly.

Abbreviation	Full form
USB	Universal Serial Bus
HDMI	High-Definition Multimedia Interface
VGA	Video Graphics Array
RJ-45	Registered Jack 45
SATA	Serial Advanced Technology Attachment
eSATA	external SATA
TRS	Tip-Ring-Sleeve
TRRS	Tip-Ring-Ring-Sleeve
HTTP	HyperText Transfer Protocol
HTTPS	HTTP Secure

Exercises

1. Distinguish a physical port from a logical port, giving one example of each.
2. Expand USB, HDMI, VGA, RJ-45, and SATA.
3. Which video port is analog and video-only, and which two common video ports are digital and also carry audio?
4. What is the difference between RJ-45 and RJ-11?
5. What is the difference between SATA and eSATA, and why is SATA not a general peripheral port?
6. Which port is commonly used to connect a monitor and supports high refresh rates and multiple monitors, and which connector is used for a wired Ethernet connection?
7. Explain the difference between TRS and TRRS audio plugs.
8. Name two features of USB that make attaching devices convenient.

Solutions

1. A physical port is a hardware socket such as a USB port or an RJ-45 port. A logical port is a software number that identifies a network service, such as HTTP port 80.
2. USB = Universal Serial Bus; HDMI = High-Definition Multimedia Interface; VGA = Video Graphics Array; RJ-45 = Registered Jack 45; SATA = Serial Advanced Technology Attachment.
3. VGA is the analog, video-only port. HDMI and DisplayPort are digital and carry both video and audio.
4. RJ-45 is the eight-pin connector used for wired Ethernet networking; RJ-11 is the smaller telephone jack.
5. SATA connects internal drives (HDD, SSD, optical drive) to the motherboard. eSATA is external SATA, used for an external drive. SATA is a storage interface — it attaches drives — so it is not a general-purpose peripheral port such as USB.
6. DisplayPort is the monitor port for high refresh rates and multiple monitors; RJ-45 is the connector for a wired Ethernet connection.
7. TRS (Tip-Ring-Sleeve) has three contact areas and carries stereo audio; TRRS (Tip-Ring-Ring-Sleeve) has four and adds a microphone channel.

8. USB supports plug and play and hot swapping, and it carries both data and power over the same connection.